Mixed Numbers & Improper Fractions

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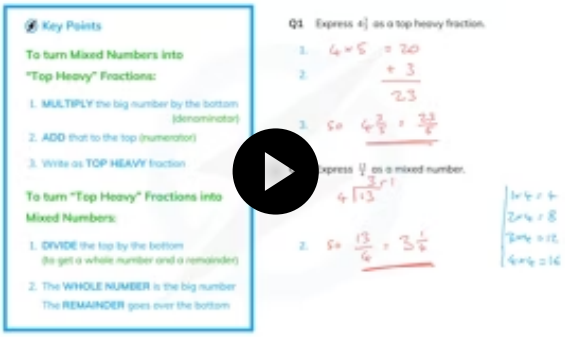

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Mixed numbers & improper fractions

What are mixed numbers & improper fractions?

- A **mixed number** has an **integer** part and a **fraction** part
  - $3\frac{3}{4}$ has the whole number 3 and the fraction $\frac{3}{4}$, meaning "three and three quarters"
- An **improper fraction** is also known as a **top-heavy fraction**
- An **improper fraction** is a fraction where the **numerator** is **bigger** than the **denominator**
  - $\frac{15}{4}$ means "fifteen quarters"
- Any fraction that is **greater than 1** can be written as either a **mixed number** or an equivalent **improper fraction**

How do I convert a mixed number into an improper fraction?

- Consider $5\frac{2}{7}$

Help me understand

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

Therefore, 5 is equal to 5 × 7 = 35 sevenths

- **STEP 2**
  Add the **numerator** of the fraction part
  - 35 sevenths and 2 sevenths make 37 sevenths
  - $5\frac{2}{7} = \frac{37}{7}$

How do I convert an improper fraction into a mixed number?

- **Divide** the numerator by the bottom
  - For example, convert $\frac{22}{3}$ into a mixed number
  - $22 \div 3 = 7$ remainder 1
- The **integer part** of the mixed number is the **whole number**
- The **fraction part** is the **remainder** over the denominator
  - $\frac{22}{3} = 7\frac{1}{3}$

Worked Example

(a) Write $5\frac{3}{4}$ as an improper fraction.

Answer:

Split the whole number into quarters

1 whole is 4 quarters

5 wholes is 5 × 4 = 20 quarters

Add the quarters from the fraction

20 + 3 = 23 quarters

$$\frac{23}{4}$$

(b) Write $\frac{17}{5}$ as a mixed number.

Answer:

Divide the top by the bottom

17 ÷ 5 = 3 remainder 2

The final answer is 3, with 2 parts still left over

Leave the 3 as the whole number part and put the 2 over the original denominator of 5

$$3\frac{2}{5}$$

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